

Why ΔP Can Be Negative and What It Means

When you observe a **negative** $\Delta P(x)$ in your difference table, it indicates that the *conditional probability* $P(x + 1 \mid x)$ at cluster size x is *lower* than the conditional probability $P(x \mid x - 1)$ at cluster size $x - 1$. Formally, we have:

$$\Delta P(x) = \underbrace{P(x + 1 \mid x)}_{\text{new}} - \underbrace{P(x \mid x - 1)}_{\text{previous}} < 0,$$

which means

$$P(x + 1 \mid x) < P(x \mid x - 1).$$

Behavioral or Empirical Explanation

In many feeding or reward-based experiments, once a subject has collected a certain number of pellets (i.e., reached a certain cluster size), its probability of continuing to collect another pellet *drops*. This is perfectly normal:

- The probability of going from $1 \rightarrow 2$ pellets (starting a cluster) might be relatively high if the first pellet triggers further seeking behavior.
- However, the probability of going from $4 \rightarrow 5$ pellets might be lower due to satiety or diminished motivation.

Hence, a negative $\Delta P(x)$ merely signifies that $P(x + 1 \mid x)$ is smaller than $P(x \mid x - 1)$, capturing this reduced likelihood of continuing a cluster at higher sizes.

Mathematical Details

If your code compiles the conditional probabilities into a list,

$$[P(2 \mid 1), P(3 \mid 2), P(4 \mid 3), \dots],$$

then you compute the differences

$$\Delta P(x) = [P(3 \mid 2) - P(2 \mid 1), P(4 \mid 3) - P(3 \mid 2), \dots].$$

Thus, if your probabilities look like

$$[0.90, 0.80, 0.75, 0.50, \dots],$$

the consecutive differences become

$$[-0.10, -0.05, -0.25, \dots],$$

and negative values merely highlight that the conditional probability has *dropped* from one cluster size to the next.

Interpretation in Analysis

- A negative ΔP is not an error; it indicates a decrease in the likelihood of adding another pellet.
- Larger negative gaps can be especially important for identifying a “maximum drop” or “maximum gap,” which some methods use to define thresholds (e.g., between “meals” and “mega meals”).

In short, negative values are a normal part of the data’s pattern, reflecting a reduced chance of extending a cluster at higher sizes.